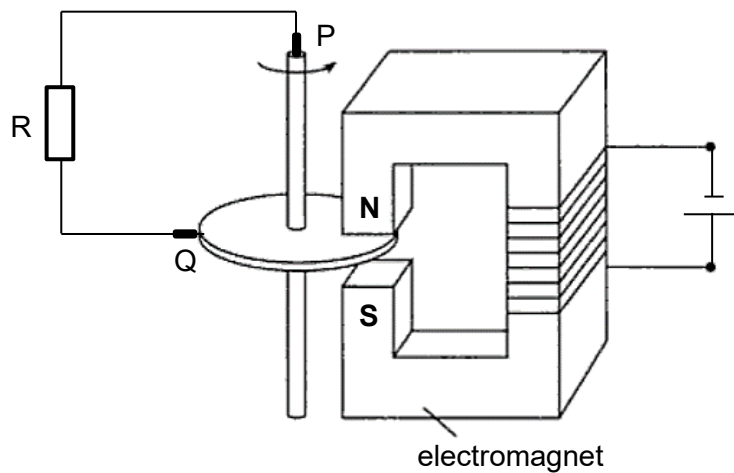


- 26** A copper disc spins freely between the poles of an electromagnet at a constant speed. P and Q are two metallic brushes making contact with the axle and the edge of the disc as shown in the diagram. A resistor R is connected across P and Q.



Which of the following statements is correct?

- | | |
|----------|---|
| A | A current flows from P through R to Q, and P is at a higher potential than Q. |
| B | A current flows from Q through R to P, and Q is at a higher potential than P. |
| C | A current flows from P to Q in the disc, and P is at a higher potential than Q. |
| D | A current flows from Q to P in the disc, and Q is at a higher potential than P. |

Answer: A

Using Fleming's RH Rule, induced current flows from the edge of the disc to the axle, i.e. Q to P.

Current then flows through the external circuit from P through R to Q and returns back to the disc.

Magnetic force causes the free electrons in the disc to drift towards the edge, so the edge becomes more negatively charged compared to the axle which becomes more electron deficient. Hence P is higher potential than Q.

[Recall also that current flows from higher to lower potential in the external circuit, but from lower to higher potential within a source of e.m.f.]

27 A sinusoidal alternating current I / A varies with time t / s according to the equation